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IoT-Based Smart Home Water Leakage Detection System Application Prototype

IT 321 - Internet of Things

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Main / Home Page

The Main / Home Page serves as the central interface of the application, allowing users to quickly monitor water usage and system status.It contains two main categories: **Dashboard** and **Activity Logs**.

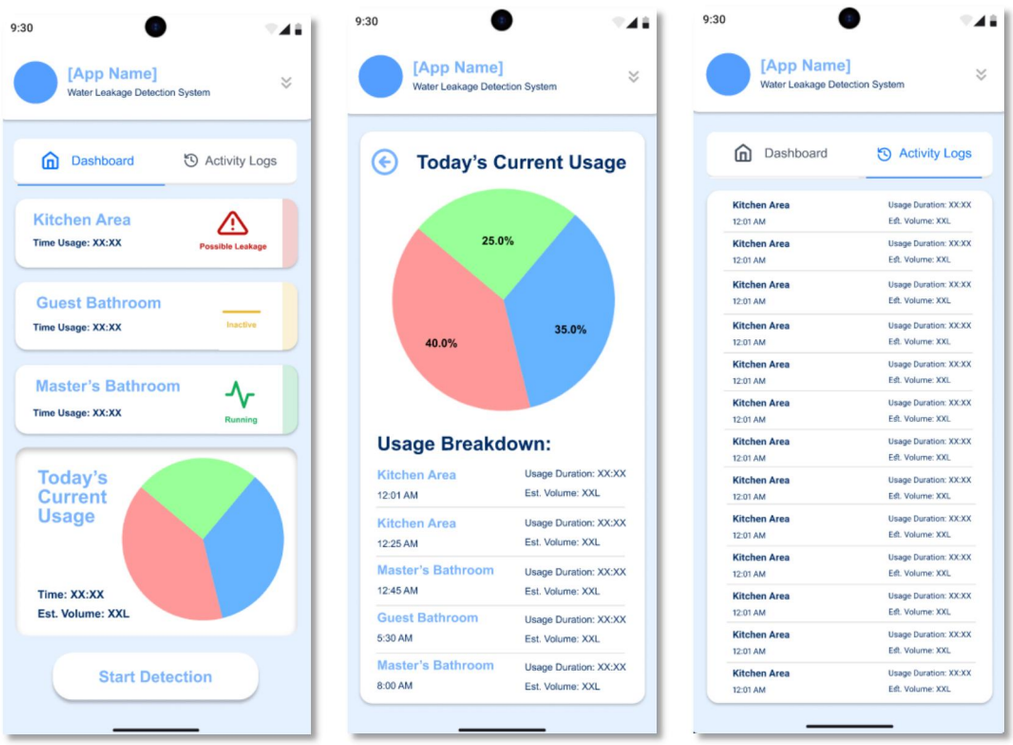
Under the **Dashboard** category, the system displays all user-defined areas (e.g., Kitchen Area, Guest Bathroom, Master's Bathroom). Each area shows **its current status** (such as Running, Inactive, or Possible Leakage) along with the **time usage or duration** indicating how long the area has been operating. Visual indicators and color cues are used to help users easily identify normal operation or potential issues at a glance.

At the bottom of the Dashboard, the **Today's Current Usage** section summarizes the user's water consumption for the day using a visual chart. This section is **clickable** and redirects the user to a more detailed **Daily Usage Record**, where specific usage breakdowns, durations, and estimated volumes per area are displayed.

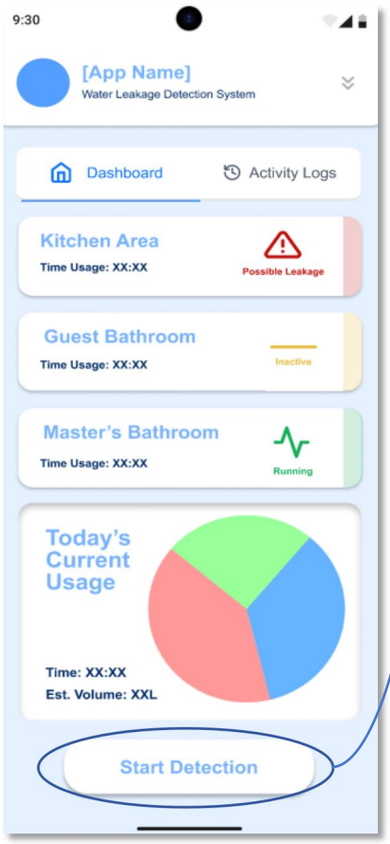
The **Activity Logs** section records the detailed history of water usage detected by the system. It provides users with a chronological list of activities based on the monitored areas they have configured. Each log entry displays the **area name** (e.g., Kitchen Area), the **time the activity occurred, the usage duration, and the estimated water volume** consumed during that session. This allows users to review past usage patterns and verify when and where water was used.

The logs are organized in a scrollable list format, making it easy for users to browse through multiple records. This feature helps users track daily usage, identify unusual activity, and support further analysis in case of suspected water leakage. Activity Logs page enhances transparency and monitoring by providing a clear and organized history of all recorded water usage events.

Overall, the Main / Home Page is designed to provide quick insights, real-time monitoring, and easy access to detailed usage information.

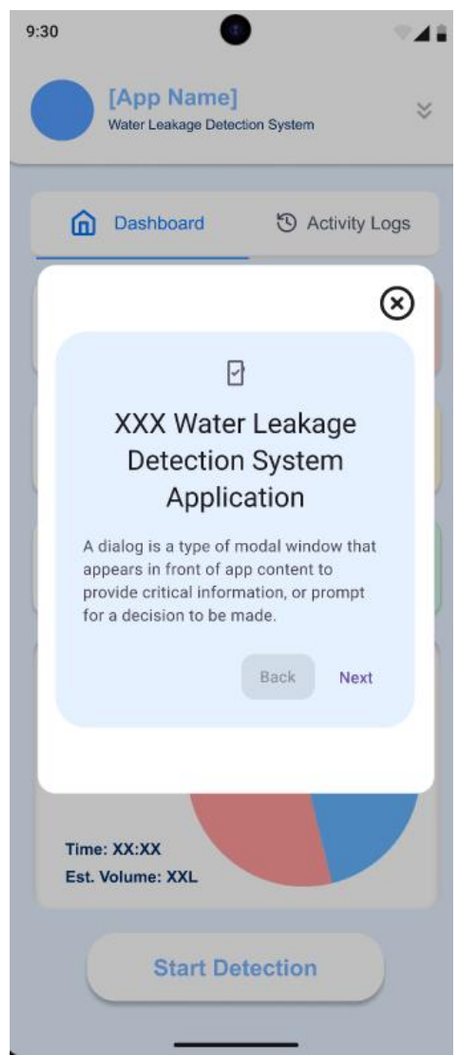


Detection Modal Flow

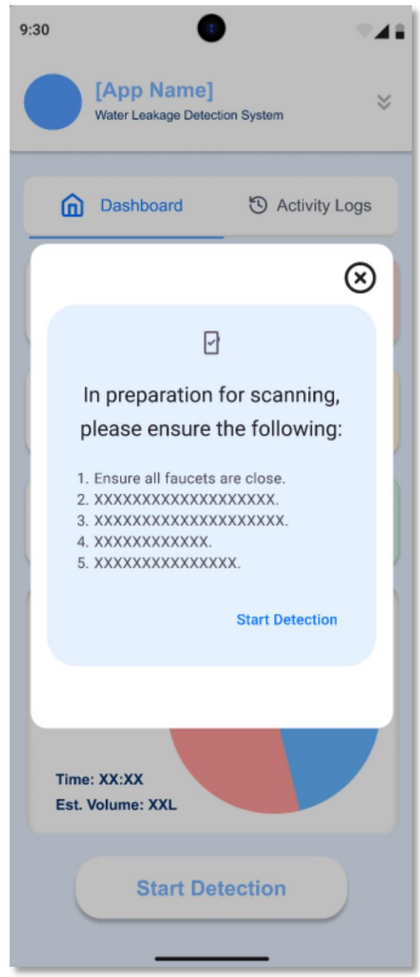


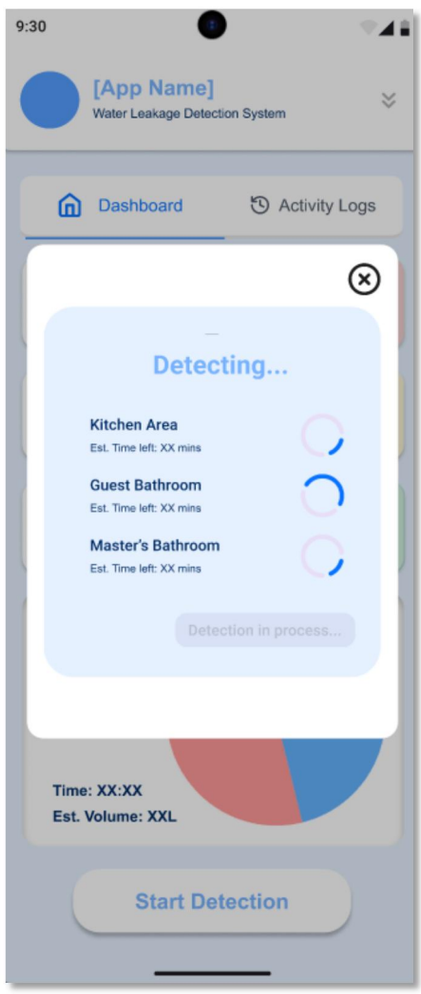
Once the user taps the **Start Detection** button on the Main / Home Page, the system initiates a guided detection process through a series of pop-up modals. These modals are designed to inform the user, ensure proper preparation, and clearly communicate the detection results.

The first modal that appears is the **Water Leakage Detection System introduction** modal. This screen provides a brief overview of the detection feature and explains that the system will analyze water usage patterns to identify potential leaks. It serves as a confirmation step, allowing users to understand the purpose of the detection before proceeding. Navigation buttons are provided so users can either return to the previous screen or continue to the next step.



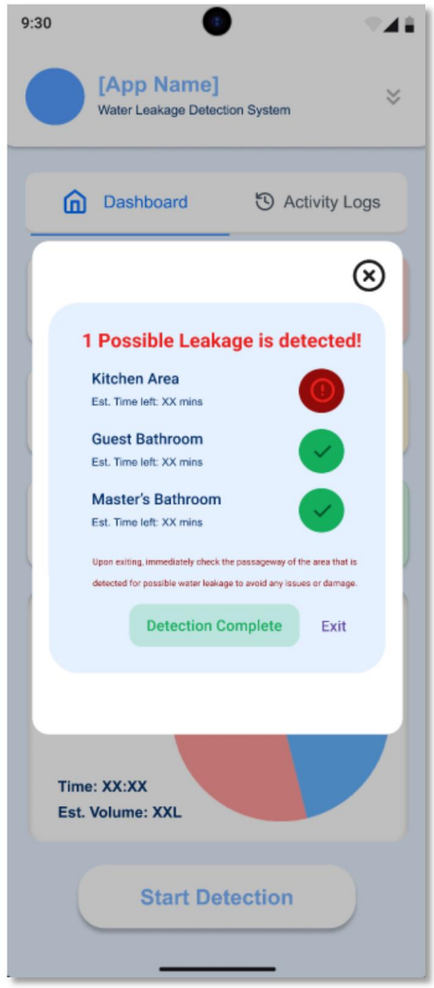
After proceeding, the **Preparation for Scanning modal** is displayed. This modal lists important instructions that the user must follow before starting the detection process, such as ensuring that no water sources are intentionally running. These instructions help prevent false detections and improve the accuracy of the scanning process. A **Start Detection button** is available to initiate the actual system scan once the user confirms that all conditions have been met.





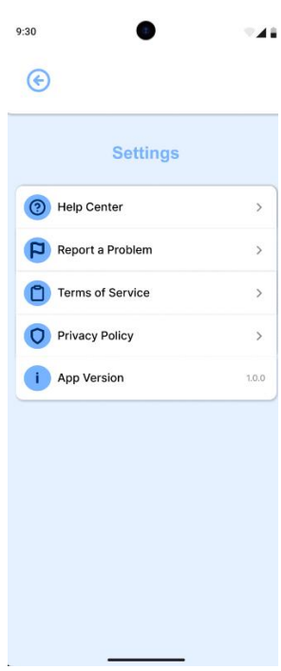
When detection begins, the **Detecting modal** appears. This screen indicates that the system is actively scanning all configured areas, such as the Kitchen Area, Guest Bathroom, and Master's Bathroom. Each area is shown with a loading indicator to represent ongoing analysis. During this phase, user interaction is limited to prevent interruptions, ensuring that the detection process runs smoothly and accurately.

Once the scan is complete, the system displays the **Detection Result modal**. This modal summarizes the outcome of the detection process by listing each monitored area along with its corresponding status. Areas with normal water usage are marked with a green indicator, while areas where abnormal or continuous water flow is detected are highlighted with a red warning indicator labeled as a possible leakage. A brief message is included to notify the user of the detected issue and to encourage further inspection or corrective action. The result modal provides a warning to the user to **complete the detection process** byreturning to the dashboard upon **exiting** the detection flow. This ensures that users can immediately take action, review results, or continue monitoring as needed.



Overall, the detection modal sequence enhances user experience by guiding users step by step through the leakage detection process. It ensures clarity, accuracy, and transparency by preparing the user, displaying real-time scanning progress, and presenting results in a clear and easily understandable manner. This design supports effective decision-making and helps users promptly identify and respond to potential water leakage issues.

[extra pages - Prototyope Completeness level: 85%]

App and Team Description Page

Application Settings Page